MILITARY SPECIFICATION

COUPLINGS AND PROPULSION CLUTCHES FOR NAVAL SHIPBOARD PROPULSION UNITS AND AUXILIARY TURBINES

1. SCOPE

1.1 Scope.- This specification covers flexible shaft couplings for propulsion units and auxiliary turbines, and clutches for turbines and reverse gears.

1.2 Classification.- Couplings and clutches shall be of the following types as specified (see 6.1).

Couplings, Shaft, Flexible

Type I - Dental (gear tooth) coupling.
Type II - Laminated disc coupling.
Type III - Rubber cushion coupling.

Clutches

Type IV - Hydraulic clutch (fluid drive).
Type V - Friction clutch, hydraulically or pneumatically actuated.
Type VI - Friction cluteh, pneumatically actuated.
Type VII - Positive drive clutch.

2. APPLICABLE DOCUMENTS

2.1 The following specifications, drawings and publications, of the issue in effect on date of invitation for bids, form a part of this specification:

SPECIFICATIONS

FEDERAL

QQ-A-561 - Aluminum Alloy 2S; Plate and Sheet.
QQ-S-691 - Steel; Plate, Marine Boiler.
ZZ-R-601 - Rubber-Goods: General Specification (Methods of Physical Tests and Chemical Analyses).
GGG-P-781 - Pullers, Bushing, Bearing, Gear, Wheel Hub and Cylinder Sleeve, Installing and Removing.

MILITARY

JAN-W-562 - Wire, Nickel-Alloy, Spring (Heat-Resistant and Age-Hardenable).
MIL-S-854 - Steel, Corrosion-Resisting: Plates, Sheets, Strips, and Structural Shapes.
MIL-B-857 - Bolts, Nuts, Studs, and Tap-Rivets (and Material for Same).
MIL-S-866 - Steel: Bars and Billets (for Carburizing).
MIL-S-871 - Steel, Molybdenum Alloys, Plates.
MIL-S-872 - Steel, Alloy, Molybdenum, Wrought.

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MIL-S-890 - Steel: Forgings and Bars for Hulls, Engines, and Ordnance (Heat Treated).
MIL-S-901 - Shockproof Equipment, Class HI (High Impact), Shipboard Application, Tests for.
MIL-C-907 - Compound, Antiseize.
MIL-D-963 - Drawings, Production (for Electrical and Mechanical Equipment for Naval Shipboard Use).
MIL-S-5000 - Steel, Chrome-Nickel-Molybdenum (4340) Bars and Reforging Stock.
MIL-A-8625 - Anodic Coatings, for Aluminum and Aluminum Alloys.
MIL-I-15024 - Identification Plates, Information Plates and Marking Information for Identification of Electrical, Electronic and Mechanical Equipment.
MIL-T-15071 - Technical Manuals for Mechanical and Electrical Equipment.
MIL-S-15083 - Steel, Castings.
MIL-R-15137 - Repair Parts for Electrical and Mechanical Equipment (Naval Shipboard Use).
MIL-C-15345 - Castings, Non-Ferrous, Centrifugal.
MIL-B-16033 - Bronze, Aluminum: Castings.
MIL-S-16113 - Steel Plate, Hull and Ordnance, Structural, Black (Uncoated) and Zinc-Coated (Galvanized).
MIL-B-16261 - Bronze, Bearing: Castings.
MIL-B-16443 - Bronze, Manganese: Castings.
MIL-A-17129 - Aluminum-Alloy Casting: Sand, Resistant to Salt-Water.
MIL-P-17286 - Preservation, Packaging, Packing, and Marking of Propulsion and Auxiliary Steam Turbines and Gears (Includes Associated Repair Parts).
MIL-G-17859 - Gear Assembly, Speed Decreaser; and Gear Assembly, Reverse (Naval Shipboard Use).

NAVY DEPARTMENT

General Specifications for Inspection of Material.
General Specifications for Machinery:
Section S1-4 - Welding and Allied Processes.
Section S45-1 - Lubrication Systems.
46M2 - Metal, Antifriction: Ingots and Castings.

DRAWINGS

BUREAU OF SHIPS

B-214 - Root Connections for Attaching Piping.
5000-S4800-3000 - Piping and Valves.

PUBLICATIONS

BUREAU OF SHIPS

NAVSHIPS 4372 - Recommendations and Proposals for Repair Parts.

(Copies of specifications, standards, drawings, and publications, required by contractors in connection with specific procurement functions should be obtained from the procuring agency or as directed by the contracting officer.)

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3. REQUIREMENTS

3.1 Materials.- Unless otherwise specified (see 6.1), or approved by the bureau or agency concerned materials as specified in table I shall be used in the construction of couplings and clutches. Material not definitely specified shall be of the quality best suited for the purpose intended.

Table I - Materials.

Part or services	Applicable specification or drawing	Remarks
Antifriction metal	46M2	Grade 2 or 3
Bearing	MIL-G-17859	
Bearing oil seals	QQ-A-561	
Bolts	MIL-B-16261	Grade II
	MIL-B-857	Class A
	MIL-S-890	Alloy No. 4
Bushing, bronze	MIL-B-16261	
Bushings, rubber	ZZ-R-601, MIL-S-890, MIL-C-15345	
Cages	MIL-S-890	Class B
	MIL-S-16113	Grade M
Cartridge sleeve	MIL-S-890	Class B
Casings, covers, and bases	MIL-S-15083	Class B or CW
	QQ-S-691	Class B
Casings, covers, and bases (small boat installation)	MIL-A-17129	Class 3 or 7
Center ring	MIL-S-890	Alloy No. 2
Clutch drive and plates	MIL-S-890, [1]	Alloy No. 2
	MIL-S-866 [1]	
Cones	MIL-S-890	Class B or Alloy No. 2
Coupling discs	MIL-S-854	
Coupling guards	MIL-S-15083	Class B
	MIL-S-16113	Grade M
	QQ-S-691	Class B
Cushions, rubber	ZZ-R-601, MIL-S-16113 QQ-S-691	Grade M Class B
Covers	QQ-S-691	Class B
Distance piece	MIL-S-890	Alloy No. 2
Drums	MIL-S-15083	Class B or CW
	QQ-S-691	Class B
	MIL-S-16113	Grade M
	MIL-S-890, MIL-S-871, MIL-S-872	
Flanges	MIL-S-890	Class B
Forgings	MIL-S-16113	Grade M
	QQ-S-691	Class B
	MIL-S-890	Class B
	MIL-A-17129	
	MIL-B-16033	

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Table I - Materials (cont'd.).

Part or services	Applicable specification or drawing	Remarks
Friction surface	46M2	Grades 2 and 3 Sintered bronze thermally bonded to steel plate; bonded asbestos composition
Hubs and sleeves	MIL-S-890 MIL-S-866[1] MIL-S-5000	Class B
Nuts	MIL-B-857	Class B
Other washers	MIL-S-890	Class C
Pins	MIL-S-866 MIL-S-5000	
Piping connection	5000-S4800-3000	
Pistons	MIL-S-890	Alloy No. 2
Plates (clutch)	MIL-S-890 MIL-S-866[1]	Alloy No. 2
Root connection	B-214	
Rotors (runners)	MIL-S-15083 QQ-S-691 47A2, MIL-A-17129, MIL-S-16113 MIL-S-890	Class B or CW Class B Grade M Alloy No. 2
Rubber (bushings and cushions)	ZZ-R-601	
Shafts	MIL-S-890	Class B or Alloy No. 2
Steel plate	MIL-S-16113 QQ-S-691 MIL-A-17129	Grade M Class B
Sleeves	MIL-S-890 MIL-S-5000[1] MIL-S-866[2]	Alloy No. 2
Spiders	MIL-S-890	Class B or Alloy No. 2
Springs	JAN-W-562	Class 2
Toothed elements	MIL-S-890	Alloy No. 2
Tubes, rubber	ZZ-R-601	
Tubes, scoap	MIL-B-16443	
Washers (spacing)	MIL-S-890	Class C

[1]Nitrided sleeves shall have a Brinell hardness of 550 minimum.
[2]Steel for carburizing, (Specification MIL-S-866), shall have physical properties as follows: Tensile strength 130,000 per square inch (p.s.i.) minimum; yield strength 90,000 p.s.i. minimum; elongation in 2 inches, 12 percent minimum hardness, minimum hardness, 58 Rockwell C.

3.1.1 The use of cast iron parts will not be permitted except where specifically approved by the bureau or agency concerned.

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3.1.2 As defined by the bureau or agency concerned, the use of critical and strategic materials shall be kept to a minimum.

3.1.3 The use of materials differing from those specified herein will be considered by the bureau or agency concerned, when the manufacturer shows the necessity for such substitution; when materials proposed are satisfactory substitutes, and when the substitute materials are readily available. The approval of substitute materials or the use of materials not covered by Military specifications does not waive the requirement that materials shall receive the usual thorough inspection required by Military specifications for materials.

3.2 Design.- The principle of reliability shall be paramount and no compromise of this principle shall be made with any other basic requirements of the design. It is the intention of this specification to obtain couplings and clutches of such a design that they will operate over a long period of years with a minimum of servicing. The design and construction shall be the lightest and most compact consistent with the following requirements listed in their order of importance.

(a) Reliability.
(b) Accessibility for repair.
(c) Efficiency.

3.3 General requirements.-

3.3.1 Telltale or revolution indicator.- A suitable type of telltale or revolution indicator, located as approved by the bureau or agency concerned, shall be provided to indicate the revolutions of the driving or driven shafts to facilitate engagement of type V, VI and VII clutches at near synchronous speed.

3.3.2 Controls.- Controls for the pneumatic or hydraulic systems of all types of clutches shall be located as approved by the bureau or agency concerned. The controls shall be of a type suitable for remote operation. The controls shall incorporate interlock features and indicating lights as approved by the bureau or agency concerned.

3.3.3 Springs.- Springs for all types of clutches shall be in accordance with class 2 of Specification JAN-W-562.

3.3.4 Securing to shaft.- Sleeves, hubs, drums, plates, flanges, rotors and spiders shall be secured to shafts or distance pieces by fitted bolts or shall be splined, shrunk or keyed as applicable. Where such parts are splined or shrunk retaining nuts shall be employed unless otherwise approved by the bureau or agency concerned. Retaining nuts shall be suitably locked against turning. Nuts used on through bolts shall be of a self locking type approved by the bureau or agency concerned.

3.3.5 Bolting.- Bolts shall be class A material and nuts class B, in accordance with Specification MIL-B-857.

3.3.6 Bearings.-

3.3.6.1 Bearings shall be designed in accordance with Specification MIL-G-17859.

3.3.6.2 All friction or rubbing surfaces shall be lined with anti-friction metal, grade 2 or 3 of Specification 46M2.

3.3.7 Bearing oil seals.- Materials for bearing oil seals shall be in accordance with table I.

3.3.8 Balance.- All coupling and clutch parts rotating at more than 150 r.p.m. shall be balanced dynamically. The addition of weights to correct for balance after finish machining will not be permitted. Balance shall be accomplished after removal of metal. Drawings submitted for approval to the bureau or agency concerned shall state permissible amount of unbalance in inch-ounces.

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3.3.9 Piping.- All piping connections shall be flanged. All piping joints shall be welded, brazed or flanged. Screwed connections shall not be used except for gages. All piping connections shall conform to Drawing 5000-S4800-3000. Root connections shall conform to Drawing B-214.

3.3.10 Welding and allied processes.- Welding, brazing and allied processes shall conform to Section S1-4 of the General Specifications for Machinery.

3.3.11 Lubrication.- Lubrication systems of couplings and clutches shall conform to Section S45-1 of General Specifications for Machinery.

3.3.12 Housing.-

3.3.12.1 Unless otherwise approved by the bureau or agency concerned, casings, covers and bases shall be constructed of cast steel, class B or CW in accordance with Specification MIL-S-15083, or shall be of welded construction incorporating cast steel, class B or CW, and steel plate in accordance with class B of Specification QQ-S-691.

3.3.12.2 For small boat installations where weight is critical, case, cover and base may be constructed of heat treated aluminum alloy, for which class 3 or 7 of Specification MIL-A-17129 shall apply.

3.3.13 Deterioration.- In order to prevent deterioration due to corrosion, all bolts, nuts, studs, pins, springs, screws, cap screws, and other fastenings or fittings used with aluminum alloy shall be of a suitable corrosion-resisting material, or of a material treated in a satisfactory manner to render it adequately resistant to corrosion. Aluminum, where used, shall be anodized in accordance with Specification MIL-A-8625 or protected against corrosion by a suitable process such as Alodine, Alrok, Bonderize, or equal. Contact between dissimilar aluminum alloys or dissimilar metals shall be avoided as much as possible. Where assembly of dissimilar metals with aluminum alloys is unavoidable, the alloys shall have their faying surfaces anodized. Where threaded surfaces or faying surfaces may require disassembly in service, an antiseize compound in accordance with Specification MIL-C-907 shall be used. For the assembly of aluminum alloy parts, through bolting is preferred using cadmium or zinc plated steel bolts and nuts; this applies with particular emphasis to comparatively high stressed assemblies or those subject to vibrating loads. Where the use of tap bolts, stud bolts, or machine screws is necessary, these shall be threaded into steel inserts cast or screwed into the aluminum alloy. The steel inserts shall be pinned or prick punched in such a manner as to prevent their backing out. All tap bolts, stud bolts, and machine screws shall be of steel with cadmium or zinc plating. No alloys of copper such as brass or bronze will be acceptable for use in threaded contact with aluminum alloys.

3.3.14 Shock requirements.- Couplings and clutches shall be designed to resist vertical, athwartship and fore-and-aft accelerations in accordance with figure 1, and are considered as part of the reduction or reverse gear with which used. Where specified by the bureau or agency concerned, equipment shall be in accordance with Specification MIL-S-901.

3.4 Type I - Dental (gear tooth) coupling.-

3.4.1 Unless otherwise specified in the contract or order dental couplings shall be of the double engagement type and shall incorporate hubs or distance pieces on which external gear teeth are machined engaging internal gear teeth machines in sleeves.

3.4.1.1 Unless otherwise approved by the bureau or agency concerned, dental coupling teeth shall be continuously supplied with lubrication by jets feeding oil to annular grooves in the coupling flanges. Ample drainage shall be provided to prevent the accumulation of sludge.

3.4.2 Where dental coupling teeth are lubricated by jets, the couplings shall be enclosed in oil tight guards. For dental couplings connecting engine and turbine shafts to pinion shafts, the guards shall terminate in flanged connections for bolting to the engine or turbine and gear housings. They shall be split horizontally. The guards shall be provided with a circumferential packed slip joint to permit axial displacement. Coupling guards shall be cast steel in accordance with class B of Specification MIL-S-15083, steel plate in accordance with grade M of Specification MIL-S-16113, or class B of Sepcification QQ-S-691.

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3.4.3 After couplings are assembled the meshing position shall be clearly marked. Provisions shall be made to restrict the fore and aft movement of the double toothed element, except that allowed for necessary working clearance.

3.4.4 Dental coupling elements shall be provided with centering lips so that centering may be accomplished independent of the coupling bolts.

3.4.5 Design shall be such that dental couplings can be disassembled without removing the connected shafts.

3.4.6 Dental couplings shall be designed with the external teeth crowned. The taper of the crown shall not exceed 0.001 inch on each side and at each end unless otherwise approved by the bureau or agency concerned. Both internal and external teeth shall be finished by shaving or grinding. It is preferred that internal teeth have the longer face.

3.4.7 Unless otherwise approved by the bureau or agency concerned toothed elements shall be made of alloy No. 2, of Specification MIL-S-890. Elements shall have a minimum Brinell hardness of 350.

3.5 Type II - laminated disc coupling. -

3.5.1 The laminated disc coupling shall consist of a stack of stainless steel discs mounted between flanges and secured thereto by means of fitted bolts.

3.5.2 Laminated disc couplings shall be arranged for single or double engagement as approved by the bureau or agency concerned. For double engagement couplings, the forward and after discs shall be connected by a forged center ring or distance piece.

3.5.3 Discs shall be secured to flanges mounted on driving and driven shafts by fitted bolts. Nuts used on coupling bolts shall be a self locking type approved by the bureau or agency concerned.

3.5.4 Unless otherwise approved by the bureau or agency concerned coupling discs shall be corrosion resisting steel conforming to Specification MIL-S-854.

3.5.5 Flanges shall be forged steel in accordance with class B of Specification MIL-S-890.

3.5.6 Center rings and distance pieces shall be forged steel conforming to alloy No. 2 of Specification MIL-S-890.

3.5.7 Spacing washers which are subject to wear shall be forged steel, alloy No. 4, of Specification MIL-S-890. Other washers shall be in accordance with class C of Specification MIL-S-890

3.6 Type III - rubber cushion coupling. - The rubber cushion coupling shall consist of rubber bushings or cushions suitably bonded to steel backs, plates or spiders and arranged to transmit power through pins or lugs.

3.6.1 Rubber cushion couplings shall incorporate rubber bushings and pins or rubber cushions of bonded sandwich construction.

3.6.2 The rubber bushings and pins or bonded cushions shall be suitable mounted in flanges or hubs. The flanges or hubs will in turn be secured to driving and driven shafts.

3.6.3 Unless otherwise approved by the bureau or agency concerned, rubber cushion couplings shall be of the double engagement type.

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3.6.4 Pin and bushing couplings shall incorporate fabricated steel cages for positioning the pins. Cages shall be of welded construction utilizing forgings conforming to class B of Specification MIL-S-890, steel plate conforming to grade M of Specification MIL-S-16113, or class B of Specification QQ-S-691. Rubber bushings shall be mounted in removable steel cartridges in a prestressed condition. The cartridge assembly shall incorporate a bronze bushing to bear on hardened steel pins. The cartridge sleeve shall be made of forged steel conforming to class B of Specification MIL-S-890. Bronze bushings shall be made from castings, conforming to Specification MIL-B-16261. Pins shall be carburized steel, conforming to Specification MIL-S-866. Pins shall be drilled and taped for pressure gun fittings for grease lubrication of bronze bushings.

3.6.5 Flanges and hubs shall be made of steel forgings conforming to class B of Specification MIL-S-890.

3.6.6 Cushions of bonded sandwich construction shall consist of suitable rubber vulcanized to a steel plate. The steel plate shall be in accordance with grade M of Specification MIL-S-16113, or class B of Specification QQ-S-691.

3.6.7 Rubber for bushings and cushions shall be in accordance with Specification ZZ-R-601.

3.6.8 Provision shall be made for positive mechanical drive in the event of malfunctioning of rubber elements.

3.7 Type IV - hydraulic clutch (fluid drive). -

3.7.1 The hydraulic clutch (fluid drive) shall consist of a driving rotor or simplified centrifugal pump runner and a driven rotor or simplified hydraulic turbine runner. Each rotor shall consist of a ring of approximately semi-circular cross section with another ring of approximately semi-circular cross section held concentrically within the first by a number of radial ribs. The two rotors together shall first form a ring of approximately circular cross section and a smaller similar one within it. The annular space between these two rings shall thus form a closed circuit or vortex ring for the circulation of the liquid when the driving rotor is rotated.

3.7.2 Primary and secondary rotors shall be of welded construction, or machined from solid forgings, or cast of aluminum alloy as approved by the bureau or agency concerned. Casings shall be steel plate or cast aluminum alloy as approved by the bureau or agency concerned. Forgings shall be in conformance with class B of Specification MIL-S-890; steel plate shall be in accordance with grade M of Specification MIL-S-16113, or class B of Specification QQ-S-691. Cast aluminum shall be in accordance with Specification MIL-A-17129.

3.7.3 Hydraulic clutches shall be provided with leakoff ports, fixed scoop tubes or sliding scoop tubes, as approved by the bureau or agency concerned for circulation of oil. Sliding scoop tubes shall be actuated hydraulically or pneumatically as approved by the bureau or agency concerned. In addition provision shall be made for manual operation in case of emergency. Scoop tubes shall be made of manganese bronze in accordance with Specification MIL-B-16443.

3.7.4 Filling of clutches shall be accomplished by admitting oil through manifolds or through a bore in pinion shaft, as approved by the bureau or agency concerned. Suitable filling and dumping valves shall be provided. The valve controls shall be arranged so that when one valve is closed the other is open. Valves shall be electrically or pneumatically actuated. In addition valves shall incorporate provision for manual operation in case of emergency.

3.7.5 Primary and secondary rotors shall be separated by an amount necessary to allow for thermal expansion and possible misalignment. The clutch may be of the twin type when necessary to reduce the overall diameter. The percentage of slip shall not exceed 3 percent unless otherwise approved by the bureau or agency concerned.

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3.7.6 The time for clutch filling and dumping shall be as specified by the bureau or agency concerned. In no case shall the dumping time exceed 3 minutes based on contract full revolutions per minute (r.p.m.).

3.7.7 Suitable means shall be provided to take the thrust developed by the clutch in both directions.

3.8 Type V - friction clutch hydraulically or pneumatically actuated. -

3.8.1 Friction clutches, hydraulically or pneumatically actuated, shall consist of flat discs or mating cone surfaces mounted on driving and driven hubs and engaged and disengaged by oil or air pressure applied to pistons.

3.8.2 Disc friction clutches shall be hydraulically or pneumatically actuated and shall operate in oil unless otherwise approved by the bureau or agency concerned. The construction shall incorporate flat discs alternately steel and sintered bronze on steel.

3.8.3 The steel discs shall be hardened to 26-36 Rockwell C. The bronze discs shall be provided with radial or spiral grooves, or both, for cooling oil circulation. The grooved and land areas of the bronze discs shall be approximately equal. Drawings submitted for approval to the bureau or agency concerned shall state chemical and physical properties of materials used for parts to which referenced purchase specifications are not applicable.

3.8.4 Cone friction clutches shall incorporate a driven member constructed of forged steel. The friction element shall be machined from a bronze centrifugal casting or shall be of suitable asbestos composition thermally bonded to a forged steel drum. Cone clutches shall operate in oil unless otherwise approved by the bureau or agency concerned. Drawings submitted for approval to the bureau or agency concerned shall state chemical and physical properties of materials used for parts to which referenced purchase specifications are not applicable.

3.8.5 Discs or cones of clutches shall be moved axially for engaging and disengaging by hydraulic or pneumatic cylinder and piston assemblies. Drawings submitted for approval to the bureau or agency concerned shall state oil or air pressure required. The steel member shall be hardened to 26-36 Rockwell C.

3.8.6 A continuous flow of oil shall be supplied from the bearing lubricating oil system to the inside of clutches, and shall be allowed to escape through clearances and grooving in order to assure proper cooling.

3.8.7 Rubbing faces of steel and bronze discs and cones shall be uniformly and smoothly finished; surface finish shall be 16 microinches r.m.s. or better.

3.8.8 Rubbing faces of all discs and cones shall be flat within 0.0015 inch and parallel within 0.002 inch.

3.8.9 Sleeves, hubs and drums shall be secured to shafts by fitted bolts or shall be splined or shrunk and keyed as applicable. Where parts are splined or shrunk on shafts, retaining nuts shall be employed unless otherwise approved by the bureau or agency concerned. Retaining nuts shall be suitably locked against turning. Nuts used on bolts shall be a self locking type approved by the bureau or agency concerned.

3.8.10 Discs and cones shall be arranged to slide axially on splines machined on shafts or sleeves as applicable. Spline teeth shall be finished to a finish of 32 microinches or better. Drawings submitted for approval to the bureau or agency concerned shall list spline tooth cutting data.

3.8.11 Disc and cone clutches shall be designed to use the same grade of oil as employed for reduction gears, engine or turbine with which used.

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3.9 Type VI - friction clutch, pneumatically actuated. -

3.9.1 Friction clutch, pneumatically actuated, shall consist of a rubber tube faced with bonded friction material arranged to engage the surface of a steel drum when inflated with compressed air.

3.9.2 Pneumatic clutches incorporating a rubber tube shall be of the dry friction type. Clutches may be arranged for either radial or axial engagement.

3.9.3 Friction linings shall be of a suitable asbestos composition. Linings may be thermally bonded to rubber tubes, secured to tubes by pins, or be of the floating plate type, as approved by the bureau or agency concerned.

3.9.4 Rubber tubes may be thermally bonded to the driving ring or secured thereto by pins, as approved by the bureau or agency concerned.

3.9.5 Unless otherwise approved by the bureau or agency concerned, the material for clutch drums and plates shall be as specified in table I.

3.9.6 Rubber tubes shall be in accordance with Specification ZZ-R-601.

3.9.7 The body of the tube shall be constructed of multiple plies of tire cord fabric reinforcement thoroughly impregnated with rubber friction compound, laminated and integrated by vulcanization. The inner tube and cover shall consist of rubber layers vulcanized to the body.

3.9.8 Each tube shall be provided with a suitable air valve vulcanized in as an integral part of the tube.

3.10 Type VII - positive drive clutch. -

3.10.1 Positive drive clutch, hydraulically actuated shall consist of a combination of gear tooth elements to transmit contract full power and friction elements for synchronizing speed of driving and driven shafts.

3.10.2 Positive drive clutches shall incorporate gear tooth elements together with friction elements. The clutch shall perform the following functions:

(a) Connect the driving and driven shafts, whose axes are essentially in line, by means of positively engaged gear teeth to transmit contract full torque of the engine or turbine, and to allow for axial displacement and misalignment of the shafts.
(b) Disconnect the driving and driven shafts so that one may rotate freely with respect to the other.
(c) Synchronize the rotary motion of the one shaft with respect to the other automatically when changing the condition of the clutch from the disconnected to the connected position before allowing the disengaged load-transmitting gear teeth to engage.

3.10.3 The clutch shall be capable of synchronizing the shafts with a maximum differential as specified (see 6.1). The torque available for synchronization shall be as specified (see 6.1).

3.10.4 The clutch shall be supplied complete with a hydraulic actuating mechanism. The clutch sleeve shall be moved axially for engaging and disengaging by means of a hydraulic cylinder and piston sets arranged with their axes parallel to the shaft axis. Each assembly shall consist of a piston yoke which is a double ended piston with a follower bearing located midway of its length to engage flanges on the sleeve.

3.10.5 The clutch sleeve shall be provided with circumferential detent grooves in combination with radially disposed loaded balls to finally position the connecting sleeve in either the engaged or disengaged position.

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3.10.6 Oil pressure for actuating the clutch mechanism shall be supplied by a motor driven pump which will be furnished as specified by the bureau or agency concerned. A hydraulic accumulator shall be furnished as approved by the bureau or agency concerned.

3.10.7 The clutch in the engaged position, shall be similar to a type I flexible coupling.

3.10.8 Hubs and sleeves shall be in accordance with Specification MIL-S-5000, nitrided, or carburized steel in accordance with Specification MIL-S-866 unless otherwise approved by the bureau or agency concerned.

3.10.9 The hub gear teeth shall have slightly crowned tooth faces. Drawings shall indicate the amount of crowning. Calculations shall be furnished with the drawings and shall provide information regarding Hertz stress at contract full power and the percent of tooth area in contact. The engaging teeth shall have chamfered ends to facilitate engagement. Teeth shall be finished by shaving or grinding.

3.10.10 The speed synchronizing elements of the clutch shall consist of a set of sintered bronze friction discs and steel discs designed to operate in oil. The requirements specified in 3.8.3 shall apply.

3.10.11 A visual indicator shall be provided which will show the position of the clutch sleeve.

3.11 Identification plates. - Identification plates shall be furnished in accordance with Specification MIL-I-15024.

3.12 Repair parts and stock components. -

3.12.1 Repair parts shall be of two general categories, namely onboard repair parts and stock repair parts. Ships Parts Control Center (SPCC), Naval Supply Depot, Mechanicsburg, Pa. is the supply demand control point (SDCP) for both categories.

3.12.2 Onboard repair parts. - Onboard repair parts shall be those clutch parts which are required to be carried onboard Naval ships for routine maintenance of parent equipment by shipboard personnel. These parts are listed in the ship's onboard allowance. Onboard repair parts shall be furnished and concurrently produced with the clutches specified herein.

3.12.3 Stock repair parts. - The procedure for determining the parts and the quantities to be purchased as stock repair parts shall be as specified in Specification MIL-R-15137; this procedure shall be followed, except that the thirty-day requirement for completion of parts II and III of NAVSHIPS Form 4372 is not applicable.

3.12.4 Stock components. - Stock components are those bureau or agency special material parts, subassemblies and assemblies which shall be required to be furnished and concurrently produced with the clutch specified herein. The bureau or agency concerned is the supply demand control point (SDCP) for stock components. Stock components shall be stocked to meet one or more of the following needs:

(a) To provide for low-rate usage replacement of clutch parts, subassemblies and assemblies incident to fleet maintenance. These parts are subject to such infrequent breakdown and slight wear that they do not qualify for inclusion as onboard repair parts but should be immediately available.
(b) To provide for replacement of clutch subassemblies and assemblies which may become inoperative due to battle damage, collision or engineering failure of such a nature and extent as to make replacement with a complete component mandatory or desirable.
(c) To provide adequate supply of bureau or agency special material to meet the mobilization reserve material requirements.

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3.12.5 Where couplings or clutches are furnished as part of propulsion gears, the stock components shall be furnished with the gears.

3.12.6 The onboard repair parts and stock components for couplings and clutches specified as follows shall be furnished by the contractor to be carried onboard.

Parts required (onboard repair parts)	Quantity to be furnished
Bearing oil seals	100 percent for one unit
Bearings (all types)	100 percent for one unit
Collar, thrust bearing	1 of each type and size
Miscellaneous wearing and operating parts	1 complete set
Oil sight flow glasses	100 percent for each ship
Shoes, thrust bearing	100 percent for one unit
Thermometers	10 percent for each ship
Thermocouple elements	100 percent for each ship
Pressure gages	10 percent for each ship
Wearing parts for relief, control and regulating valves (includes springs, diaphragms, pistons, seat inserts and bushings)	1 set for each type and size valve
Steel and sintered bronze plates for type V and VII clutches	100 percent for one unit
Cone elements	100 percent for one unit
Rubber tube, type VI clutch	1 per ship
Friction lining, type VI clutch	1 per ship
Drum, type VI clutch	1 per ship
Rubber bushings, type III clutch	100 percent
Hubs for type VII clutch	100 percent for one unit
Sleeve for type VII clutch	100 percent for one unit
Springs	100 percent for one unit
Ball detents	100 percent for one unit

3.12.7 Stock repair parts (special material) shall be in accordance with Specification MIL-G-17859 in regard to flexible couplings and fluid drive clutches.

3.12.8 To the onboard components and stock repair parts (special material) shall be added any wearing or operating parts or special gear or attachments that the bureau or agency concerned considers necessary for dependable and efficient service maintenance.

3.12.9 The manufacturer shall advise the bureau or agency concerned in detail as to the interchangeability of repair parts and stock components for couplings and clutches furnished herein and those previously furnished or being concurrently furnished under other procurements. This information shall be required prior to determination and assignment of standard Navy stock numbers to those parts.

3.13 Special tools and equipment. -

3.13.1 Special equipment shall be supplied as specified hereinafter, and shall be carried onboard.

3.13.2 One complete set of all special wrenches for the clutches, and fittings for same, in each machinery compartment shall be furnished. The set shall be arranged for convenient and accessible stowage in the compartment. Where practicable, wrenches shall be forged steel with case hardened jaws. In all cases, they shall be of the highest quality. No ordinary commerical tools need be included in such a set.

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3.13.3 Special pulling tools shall be provided where the standard pullers shown in Specification GGG-P-781 cannot be used.

3.13.4 Any other special gear or attachments that the bureau or agency concerned considers necessary for the proper maintenance of machinery in service shall be furnished.

3.14 Drawings. -

3.14.1 Drawings shall be in accordance with Specification MIL-D-963.

3.14.2 Installation drawings shall show overall dimensions and information necessary for installation. Such information shall include the following: Weights of the complete unit and major parts; foundation bolting dimensions; clearances for removal of major parts and accessory equipment. Diagrammatic sketches of the various piping systems indicating pipe sizes, types and sizes of connections, relative location of coolers, valves, filters, thermometers, thermocouples and gages, recommended by the contractor, shall be forwarded to the bureau or agency concerned and activities as directed for comments and use in preparation of piping arrangement drawings. For ready reference, piping diagrams shall show the pump capacities, normal working pressures, and flow in gallons per minute through the piping systems. Diagrams of electrical circuits shall be furnished as applicable.

3.15 Instruction books. - Instruction books shall conform to type B of Specification MIL-T-15071. Unless otherwise specified by the bureau or agency concerned, coupling and clutch instructions shall be incorporated with the gear, engine, turbine, generator, pump or compressor instruction book.

3.16 Workmanship. - All work shall be executed in a thorough, workmanlike and substantial manner. Unless otherwise specified in the contract or order, the Government inspector having jurisdiction at the manufacturer's plant shall be the sole judge of whether or not work is satisfactory and acceptable to the bureau or agency concerned.

4. QUALITY ASSURANCE PROVISIONS

4.1 Inspection. -

4.1.1 All materials used in couplings and clutches shall be subjected to the inspection requirements of the closest equivalent and the most nearly applicable Military specification covering similar material.

4.1.2 All gear teeth of couplings and clutches, and all shafts shall be subjected to inspection by the magnetic powder method.

4.1.3 Welded parts of couplings and clutches which are subject to dynamic loads shall be radiographed in such areas as may be directed by the bureau or agency concerned.

4.1.4 Special attention shall be given to the thorough cleanliness of all parts of the lubricating system, and care shall be taken to insure that full flow can be circulated throughout the entire system.

4.2 Testing. -

4.2.1 Couplings and clutches shall be tested with the reduction gear, engine or turbine with which used at contract full power at contract designed r.p.m. of propeller.

4.2.2 Clutches, types V, VI and VII, shall be operated for a period of thirty minutes at a speed of twenty percent above maximum designed speed. The test may be conducted with or without the reduction gear, turbine or engine connected and with or without load.

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4.2.3 During the tests specified in 4.2.1 and 4.2.2 a check shall be made of the couplings or clutches including the following: adequate lubrication at all speeds, quiet running, freedom from vibration and excessive heating, wearing of parts, tightness of casing and piping connection and adequacy of bearing oil seals. Any unsatisfactory conditions or evidence of probable unsatisfactory operation in service, apparent on the tests shall be considered cause for rejection.

4.3 Inspection procedures. - For Naval purchases, the general inspection procedures shall be in accordance with General Specifications for Inspection of Material.

5. PREPARATION FOR DELIVERY

5.1 Couplings, clutches, stock components and repair parts shall be preserved, packaged, packed and marked for the level of shipment as specified (see 6.1) in accordance with Specification MIL-P-17286.

6. NOTES

6.1 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification.
(b) Type required (see 1.2).
(c) Materials required (see table I).
(d) Synchronizing torque and speed (see 3.10.3).
(e) Whether for domestic shipment - immediate use; domestic shipment and storage; or overseas shipment (see 5.1).

Patent notice. - When Government drawings, specifications, or other data are used for any purpose other than in connection with a definitely related Government procurement operation, the United States Government thereby incurs no responsibility nor any obligation whatsoever; and the fact that the Government may have formulated, furnished, or in any way supplied the said drawings, specifications, or other data is not to be regarded by implication or otherwise as in any manner licensing the holder or any other person or corporation, or conveying any rights or permission to manufacture, use, or sell any patented invention that may in any way be related thereto.

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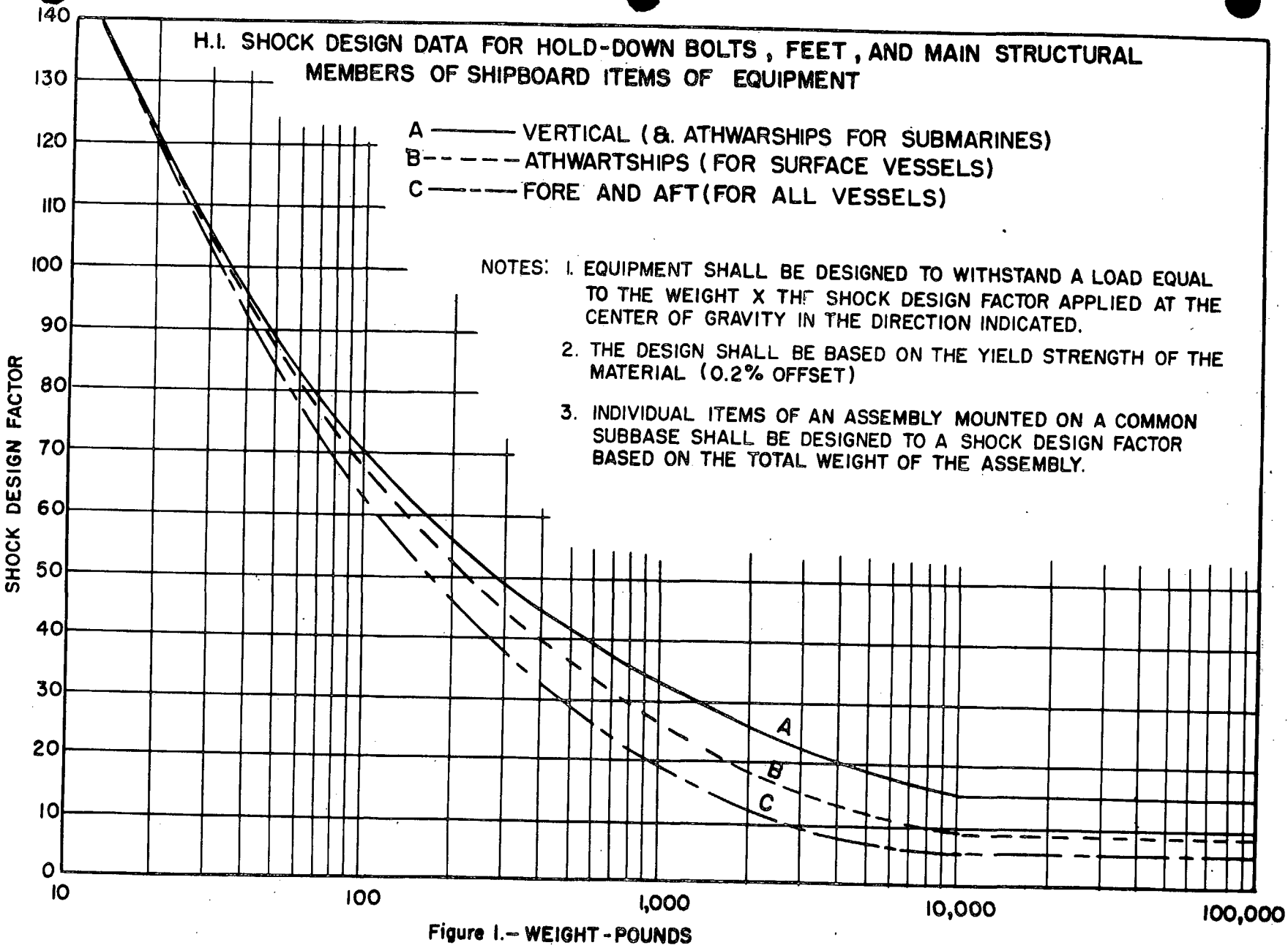


Figure 1.– WEIGHT - POUNDS

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